Precambrian Research 329 (2019) 211–223

Contents lists available at ScienceDirect

Precambrian Research

journal homepage: www.elsevier.com/locate/precamres

Palaeomagnetism of the 1.89 Ga Boonadgin dykes of the Yilgarn Craton: Possible connection with India

Yebo Liu[a,b,*], Zheng-Xiang Li[a,b], Sergei Pisarevsky[a,b], Uwe Kirscher[a,b], Ross N. Mitchell[a,b], J. Camilla Stark[a,b]

[a] ARC Centre of Excellence for Core to Crust Fluid Systems (CCFS), Curtin University, GPO Box U1987, Perth, WA 6845, Australia
[b] Earth Dynamics Research Group, The Institute for Geoscience Research (TIGeR), School of Earth and Planetary Sciences, Curtin University, GPO Box U1987, Perth, WA 6845, Australia

A B S T R A C T

A palaeomagnetic study was carried out on the newly identified 1888 ± 9 Ma Boonadgin dyke swarm of the Yilgarn Craton in Western Australia. The Bonnadgin dykes yield a mean direction of magnetisation of $D = 143^{\circ}$, $I = 13^{\circ}$, $k = 37$ and $\alpha_{95} = 8^{\circ}$, based on samples from 10 diabase dykes, with a corresponding palaeopole at 47° S, 235° E, $A_{95} = 6^{\circ}$. A positive baked contact test establishes the primary nature of the magnetisation. The ca. 1.89 Ga palaeopole suggests that the Yilgarn Craton was near the equator at this time, and the Boonadgin dyke swarm can be interpreted to represent an arm of a radiating dyke swarm that shared the same plume centre with coeval mafic dykes in the Dharwar and Bastar cratons of southern India. We therefore propose that the West Australian Craton (WAC, consisting of the Yilgarn and Pilbara cratons) and the South Indian Block (SIB, consisting of the Dharwar, Bastar, and Singhbhum cratons) were connected ca. 1.89 Ga. Globally, available high-quality palaeopoles of similar age allow the West Australian Craton to be placed northwest of proto-Laurentia during the assembly of the supercontinent Nuna.

1. Introduction

The lack of high-quality Palaeoproterozoic palaeomagnetic poles for most cratons presently hampers the debate over the assembly and configuration of Palaeoproterozoic-Mesoproterozoic supercontinent Nuna, of which the West Australian craton (WAC) is considered to be a crucial part (Belica et al., 2014; Betts et al., 2016; Evans et al., 2016; Evans and Mitchell, 2011; Klein et al., 2016; Meert et al., 2011; Pehrsson et al., 2016; Pisarevsky et al., 2014a; Zhang et al., 2012).

Mafic dykes represent ideal targets for palaeomagnetic studies as they are strongly magnetized and they are also routinely datable with the advent of U-Pb geochronology on baddeleyite. Ubiquitous ca. 1.89 Ga mafic magmatism is found on most Precambrian cratons, from which palaeomagnetic studies have yielded a series of reliable palaeopoles over the past decade (Belica et al., 2014; Buchan et al., 2016; Kilian et al., 2016; Klein et al., 2016; Letts et al., 2011). Combining palaeomagnetic constraints and matching the geometry of coeval dyke swarms has been demonstrated to be an effective way to reconstruct the configurations of two or more continents (Bleeker and Ernst, 2006; Ernst et al., 2010). This approach, however, was not applicable to the WAC ca. 1.89 Ga until the recent identification of the 1888 ± 9 Ma Boonadgin dyke swarm (Stark et al., 2017).

In this study, we report new palaeomagnetic data from the Boonadgin dyke swarm. By comparing our new results with existing data from other continents, we explore global palaeogeography ca. 1.89 Ga, particularly in the neighbourhood of the WAC.

2. Regional geology and previous work

The Yilgarn Craton is the largest Archaean craton in Australia, assembled between ~2940 Ma and 2650 Ma through the accretion of a series of terranes with a general eastward younging trend (Chen et al., 2003; Myers, 1993). The Yilgarn Craton is bound by the Palaeoproterozoic Capricorn Orogen to the north, the late Mesoproterozoic to Neoproterozoic Pinjarra Orogen to the west, and the late Palaeoproterozoic to Mesoproterozoic Albany-Fraser Orogen to the south and southeast (Johnson et al., 2011; Myers, 1993; Myers et al., 1996) (Fig. 1). The craton is composed mainly of metasediments, metavolcanics, granites, and granitic gneiss that formed between 3000 and 2600 Ma (Myers, 1993; Pidgeon, 1990; Wilde et al., 1996). The Yilgarn Craton collided with the Pilboyne Craton (combination of the Pilbara Craton and the Glenburgh Terrane of the Gascoyne Province) during the

* Corresponding author at: School of Earth and Planetary Sciences, Curtin University, GPO Box U1987, Perth, WA 6845, Australia.
E-mail address: yebo.liu@postgrad.curtin.edu.au (Y. Liu).

https://doi.org/10.1016/j.precamres.2018.05.021
Received 11 August 2017; Received in revised form 15 May 2018; Accepted 15 May 2018
Available online 16 May 2018
0301-9268/ © 2018 Elsevier B.V. All rights reserved.

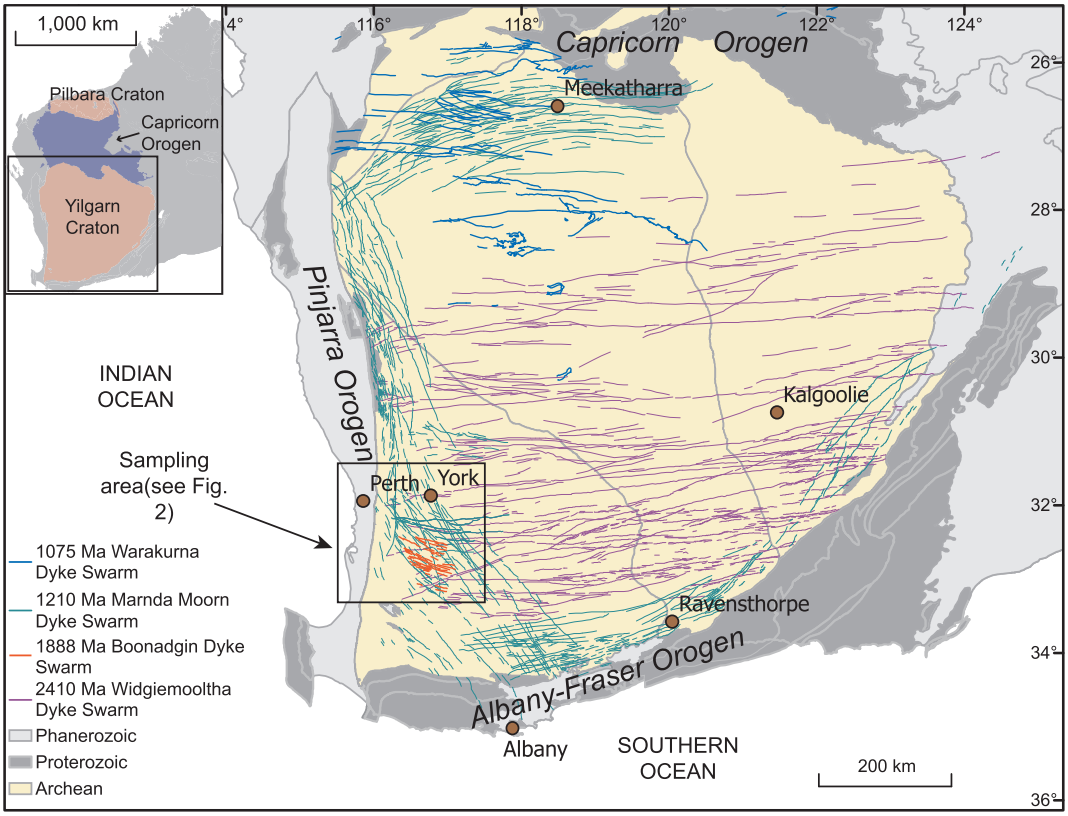


Fig. 1. Simplified geological map showing major dyke swarms in the Yilgarn Craton. The inset shows the location of the Yilgarn and Pilbara cratons within Western Australia. The dykes are mapped based on 1:2.5 M Geological Map of Western Australia 2015 published by the Geological Survey of Western Australia (https://dasc.dmp.wa.gov.au/dasc/).

2005–1950 Ma Glenburgh Orogeny (Johnson et al., 2013, 2011; Sheppard et al., 2010), thus forming the West Australian Craton.

Numerous mafic dyke swarms intrude the Yilgarn Craton (Fig. 1), among which three distinct swarms have been well-recognized and palaeomagnetically studied: the ~2410 Ma Widgiemooltha dyke swarm (Evans, 1968; Smirnov et al., 2013), the ~1210 Ma Marnda Moorn dyke swarm (including the Muggamurra, Boyagin, Wheatbelt and Gnowangerup-Fraser dykes; see Pisarevsky et al., 2003, 2014b; Wang et al., 2014; Wingate and Pidgeon, 2005 and references therein), and the ~1075 Ma Warakurna dyke swarm (Wingate et al., 2004, 2002).

Apart from those three well-known swarms, many dykes in the Yilgarn Craton remain unclassified. The southwestern part of the Yilgarn Craton, where this study was carried out, has a particularly dense network of dykes with various trends, where only the Marnda Moorn and the Widgiemooltha dykes have been previously identified (Fig. 2). The Widgiemooltha dykes are generally easy to distinguish from others by their distinct ENE-WSW trends and aeromagnetic characteristics. Whereas dyke swarms in this region can rarely be traced for more than a few kilometres, the Widgiemooltha dyke swarm can be followed, in outcrop or aeromagnetically, for up to 600 km. The orientation of the Marnda Moorn dykes in this area varies widely from E-W to N-S, but with a prevailing NW-SE trend. Due to this observation, all broadly NW-SE-trending dykes in this area were conventionally classified as Marnda Moorn dykes (Boyd and Tucker, 1990; Lewis, 1994; Spaggiari et al., 2009; Tucker and Boyd, 1987). However, a recent TIMS and in situ SHRIMP U-Pb geochronological study (Stark et al., 2017) has led to the recognition of a new dolerite dyke swarm in this area with a broadly WNW-ESE trend. The 1888 ± 9 Ma "Boonadgin" dyke swarm and is the target of the present study (Fig. 2).

Giddings (1976) studied 54 dykes along the western margin of the Yilgarn Craton, including 49 dykes from the Perth region and 5 dykes from the Ravenstrope region (Fig. 1), aiming to establish an age chronology of the dykes palaeomagnetically. While he identified five distinct groups of palaeomagnetic directions, the Rb-Sr ages implied that there were at least six, possibly seven periods of dyke emplacement. The generations of dykes in the Perth region were named "YA–YF", where baked contact tests were performed at the YB and YC groups. Giddings (1976) considered these contact tests to be positive and thus regarded the magnetic remanence as primary. In contrast, Halls and Wingate (2001) performed a more extensive baked contact test for the YB dykes and re-interpreted the results as a secondary, remagnetisation remanence. Those authors attributed the difference between the two studies to the fact that Giddings (1976) only obtained samples from the weathered surface of the unbaked zone, thus resulting the misleading positive contact test. Halls and Wingate (2001) concluded that the YB dykes were remagnetised, possibly in Mesozoic time. Giddings (1976) also suggested that the YA group of dykes and the Ravenstrope dykes, which have been dated at that time by Rb-Sr method at 2500 ± 100 Ma, could be similar in age based on their similar magnetic directions. However, the Ravensthorpe dykes have been re-dated by U-Pb geochronology and are now attributed to the ca. 1210 Ma Marnda Moorn LIP (e.g. Wingate and Pidgeon, 2005). Pisarevsky et al. (2003; 2014a,b) reported a primary remanence for these dykes, supported by a positive baked contact test. Evans (1968) and Smirnov et al. (2013) reported robust palaeomagnetic data with positive baked contact tests for the 2.41 Ga Widgiemooltha dykes that included samples from the present study area. Pisarevsky et al. (2015) published palaeomagnetic data for the ca. 2.40 Ga Erayinia dykes. The Erayinia pole plots close, but not identical, to the Widgiemooltha pole, which is consistent with their slight age difference.

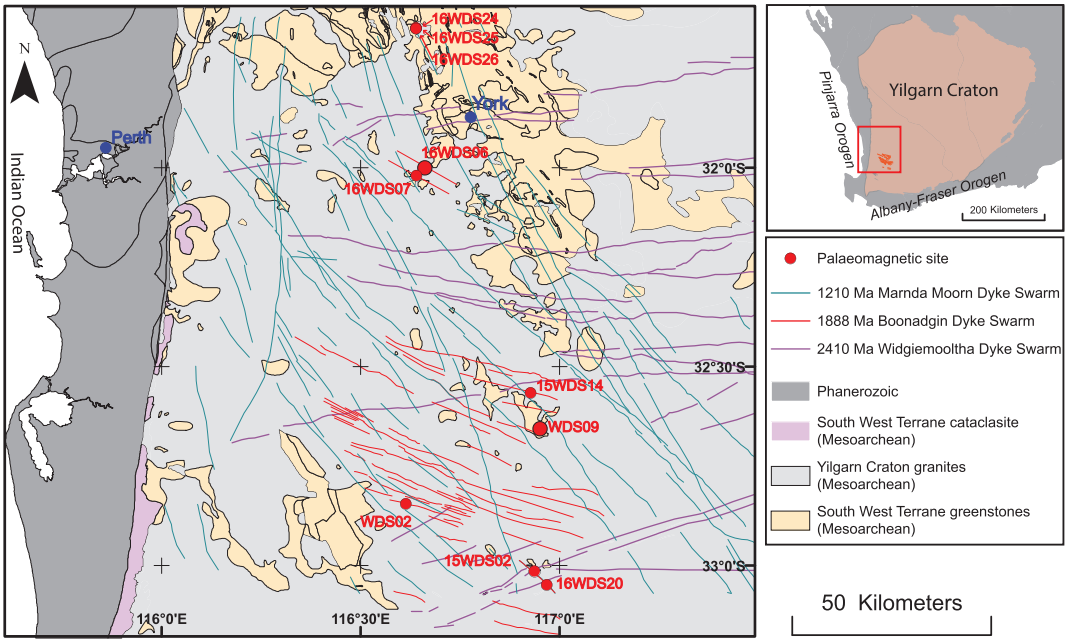


Fig. 2. Simplified geological map of the sampling area. Enlarged red solid circles represent sample sites with U-Pb dating.

3. Methods

A total of 96 cores from 10 sites were collected for palaeomagnetic analysis (Fig. 2). Each site represents a distinct dyke. 15WDS02 and 16WDS20 might appear to be in line and linkable; however, they should represent two different dykes as they revealed different magnetic signals (see the results section for details). A minimum of 6 samples (usually 8–12; Table 1) of standard 24 mm diameter were obtained from each site using a gasoline-powered portable drill with a water-cooled diamond drill bit. In addition, whenever the chilled contact was visible, the surrounding baked and unbaked country rocks were both sampled for the purpose of performing baked contact tests (sites WDS09, 15WDS02, and 16WDS24). Each core was oriented using a magnetic compass, combined with a sun compass whenever possible. As finer-grained parts of a dyke are more suitable for palaeomagnetic analysis, special efforts have been made to try to identify and sample the finest-grained parts available, ideally targeting the chilled margins of each dyke. We also tried to avoid sampling at topographically elevated points like ridges since the magnetic remanences at such outcrops are more likely to be affected by lightning strikes.

Prior to any other experiments, anisotropy of magnetic susceptibility (AMS) and bulk magnetic susceptibility (MS) were measured for all specimens using an AGICO MFK1 Kappabridge. All susceptibility measurements have been analysed and plotted with the Anisoft

Table 1
Palaeomagnetic results from the Boonadgin dyke swarm and the host rocks.

Site (dyke)	Trending (°)	Width (m)	N/n	Slat. (°S)	Slong. (°E)	Decl. (°)	Incl. (°)	k	α_{95} (°)	Plat. (°N)	Plong. (°E)	Dp (°)	Dm (°)
"Normal" polarity													
15WDS02	304	~7	9/5	33.013933	116.936400	153.5	6.2	52.2	10.7	−51.2	251.7	5.4	10.7
15WDS14	321	~40	7/3	32.578117	116.923983	120.6	0	46.8	18.2	−25.4	224.6	9.1	18.2
16WDS06*	300	~35	12/5	31.999554	116.661655	148.8	4.3	33.5	13.4	−48.1	245.8	6.7	13.4
16WDS07	325	~45	13/6	32.020191	116.639917	142.7	15.5	27.6	13.0	−47.8	233.3	6.9	13.4
Mean of "normal" polarity		4			141.5	6.7	26.1	18.3	−43.7	237.5	9.2	18.4	
"Reverse" polarity													
WDS02	304	~50	6/4	32.844973	116.613044	332.6	−21.5	23.9	19.2	−56.8	241.1	10.7	20.2
WDS09*	307	~2	7/3	32.655888	116.950544	331.6	−27.2	25.4	25.0	−58.4	235.5	14.8	27.2
16WDS20	303	~20	12/6	33.048898	116.967280	314.4	−20.8	38.2	10.0	−42.7	224.3	5.5	10.5
16WDS24	312	10.1	13/13	31.649379	116.638855	317.8	−6.8	22.1	8.6	−41.4	233.3	4.3	8.6
16WDS25	315	13.1	9/7	31.648924	116.638975	323.6	−9.4	57.0	8.6	−48.6	235.8	4.5	8.8
16WDS26	315	0.55	8/7	31.650388	116.638680	320.4	−18.8	47.7	8.2	−47.3	228.8	4.4	8.5
Mean of "reverse" polarity		6			323.3	−17.5	60.3	8.7	−48.8	232.9	4.7	9.0	
10 dykes combined		**10**			**142.5**	**13.2**	**37.3**	**8.0**	**−46.8**	**234.9**	**4.2**	**8.2**	
Baked-contact test of 16WDS24 (dyke width 10.1 m)													
Baked zone (< 10.1 m)[a]		8/5	31.649379	116.638855	323.0	-20.6	18.1	16.2	-49.9	229.8	8.9	17.0	
Unbaked zone (> 10.1 m)[b]		17/6	31.649379	116.638855	4.2	-43.9	33.5	11.7	-83.0	329.6	9.1	14.6	

N/n = number of demagnetised/used samples; Slat, Slong = sample locality latitude and longitude; Decl, Incl = site mean declination, inclination; k = precision parameter of Fisher (1953); α_{95} = semi-angle of the 95% cone of confidence; Plat, Plong = latitude, longitude of the palaeopole; Dp, Dm = semi-axes of the cone of confidence about the pole at the 95% probability level.

[a] Gneiss samples collected within 10.1 m from the western margin of 16WDS24. Mean calculation includes the low temperature component of sample 16WDS24-9 (for details see text and Fig. 7).

[b] Gneiss samples collected further than 10.1 m from the western margin of 16WDS24. Mean calculation includes the high temperature component of sample 16WDS24-9 (for details see text and Fig. 7).

* Sites with U-Pb dating.

software (Chadima and Jelínek, 2008). In order to determine the magnetic mineralogy of the dykes, representative specimens were magnetised along three orthogonal axes using magnetic fields of 3 T, 0.4 T and 0.12 T, respectively (Lowrie, 1990), using a 2G MMPM9 pulse magnetiser. The isothermal remanent magnetisations (IRMs) were then subjected to thermal demagnetisation in 14 to 20 steps from 50 °C to 610 °C. K-T curves (magnetic susceptibility versus temperature) were obtained using an AGICO MFK1 Kappabridge with a CS4 furnace in Ar-atmosphere.

After measuring the natural remanent magnetisation (NRM) of all samples, at least one specimen per sample has been subjected to stepwise AF demagnetisation up to 100 mT and/or thermal demagnetisation up to ~590 °C. In order to limit the effect of lightning-induced secondary remanence residing in single domain magnetite, the samples were occasionally first AF demagnetised up to 20mT and subsequently thermally demagnetised up to ~600 °C. After each step, the magnetisation was measured using an AGICO JR 6A spinner magnetometer. An average of ~12 demagnetisation steps were used to isolate remanence components. AF and thermal demagnetisations were carried out using a Molspin AF demagnetiser and Magnetic Measurements Ltd thermal demagnetisers, respectively. All demagnetisation and subsequent experiments, except for the K-T analyses, were carried out in either a low-field coiled room at the University of Western Australia, or in a magnetically shield room at Curtin University, both part of the Western Australian Palaeomagnetic and Rock Magnetic Facility located in Perth. The K-T analyses were conducted in laboratory of environmental magnetism of Guangzhou Institute of Geochemistry.

Mean directions of individual components of the magnetic remanence of each specimen were calculated using principal component analysis (Kirschvink, 1980). Site-mean directions were calculated using Fisher statistics (Fisher, 1953). Calculations were performed using the Remasoft software (Chadima and Hrouda, 2006). The GPlates program (www.gplates.org) was used for palaeogeographic reconstructions.

4. Results

4.1. Rock magnetism

The K-T curves show Curie temperatures between 580 and 600 °C, suggesting that low-titanium titanomagnetite or pure magnetite are the main magnetic carriers. An increase of magnetic susceptibility just before the Curie temperature (the Hopkinson peak, Dunlop and Ozdemir, 1997)are observable (Fig. 3), indicating the presence of SD and PSD (titano)magnetite. The heating and cooling curves of sample 15WDS2F1 are different, indicating that mineral phase changes occurred during the heating process.

The demagnetisation of a composite three axis IRM (Lowrie, 1990) for the Boonadgin dykes shows a general dominance of the soft coercivity fraction in all samples, which is removed between 550 and 590 °C, indicating MD (titano)magnetite being the dominant carrier of the magnetic remanence. While the 0.12 T fraction is most prominent, the 0.4 T fraction is always present and shows a similar demagnetisation behaviour (Fig. 3) confirming that SD/PSD (titano)magnetite are present in the samples. Occasionally, a sharp drop of magnetic intensity is visible at ~300 °C (Fig. 3), which could be attributed to the inversion of maghemite to hematite or the presence of pyrrhotite (Dunlop and Ozdemir, 1997). It should be noted that pyrrhotite is converted to magnetite during heating, which is often accompanied by an increase in the magnetic intensity but not seen in this study (Fig. 3).

The results of rock magnetic experiments demonstrate that SD/PSD (titano)magnetite are consistently present in the samples. We therefore consider the Boonadgin dykes are capable of preserving hard magnetic remanence.

4.2. Anisotropy of magnetic susceptibility (AMS)

The degree of AMS ($P = K_{max}/K_{min}$) for all studied samples is generally low (< 1.10; Fig. 4a) with only a few exceptions. The anisotropy ellipsoids are predominantly strongly prolate except for a few marginally to moderately oblate cases (Fig. 4b). The low degree of anisotropy is typical of mafic dykes (Chadima et al., 2009).

Among the 10 sites we used to calculate the palaeomagnetic pole, 15WDS02 and 16WDS20 do not show observable magnetic fabrics and are excluded from further AMS analysis. The remaining sites show either normal (K_{max} axes are in the plane of dykes; Fig. 4c) or inverse (K_{max} axes are normal to the plane of dykes; Fig. 4d) AMS fabrics.

In the cases of normal AMS fabrics, the magnetic lineation (i.e., the clustered direction of K_{max}) is generally considered to represent the magma flow direction (Knight and Walker, 1988). In the five dykes with normal fabrics (Fig. 4c), the inclinations of the K_{max} axes are low, indicating horizontal to subhorizontal flow patterns.

The inverse AMS fabric, which appeared in three of the studied dykes (Fig. 4d), has been frequently observed, but is not well-understood (Canon-Tapia, 2004; Chadima et al., 2009). Several explanations have been proposed: (i) the single domain effect (Potter and Stephenson, 1988), which probably does not apply to this study because the rock magnetic experiments suggest multi-domain magnetite as the main magnetic phase (Fig. 3); (ii) post-emplacement alteration (Canon-Tapia, 2004); (iii) elongate particles could roll when their long axes are normal to the flow directions (Jeffery, 1922). Without further analysis, the reason causing inverse fabrics in this study remains inconclusive.

There are cases where the magnetic fabrics of some samples do not agree with the overall fabric of the dyke. We suspect that this is because the samples were from loose boulders. The samples with incompatible magnetic fabrics also have inconsistent palaeomagnetic directions compared to the directions of other samples from the same dyke. We therefore excluded both the AMS and palaeomagnetic results from such suspected loose boulders.

4.3. Palaeomagnetism

Stepwise demagnetisation revealed high-temperature components decaying toward the origin for nearly all samples (Fig. 5). We noticed that within-site scatter of several sites is quite large, which is ascribed to two possible reasons. First, the outcrop condition in the sampling region is relatively poor. Due to prolonged lateritic weathering, few dykes in this area present continuous outcrops. Among the ten dykes we studied, three dykes (16WDS24, 16WDS25, 16WDS26) were sampled from fresh road-cuts and others are from field exposures. Dykes exhibiting significant within-site scatter were sampled from small linear field outcrops and some of those could be slightly dislocated (suspected boulders). Second, a possibility of remagnetisation by lightning strikes cannot be excluded in the topographically flat Yilgarn Craton, as some of these rocks reveal rather high Koenigsberger ratios (> 20). However, we identified well-clustered high-temperature components from fresh outcrops such as road-cuts without any indications of blocks being not in situ, which do not show systematic deviations compared to the remaining sites and give confidence to our overall conclusions (Fig. 5 and Fig. 6).

Bearing these two reasons in mind, we excluded results from samples collected from suspected loose boulders (based on field observations and AMS data), or from those probably affected by lightning strikes. The demagnetisation of remaining samples from 10 sites revealed two remanence components: a single polarity, low temperature component (CL) and a dual-polarity, high temperature component (CH), based on their unblocking behaviour (Fig. 6a and b).

The CL component has been isolated in three dykes (16WDS24 – 16WDS26), generally < 370 °C. It has a northern steep upward direction (D = 349.8°, I = −66.9°, α_{95} = 17.5°), which is close to the present-day Earth magnetic field direction in this region (D = 358.7°,

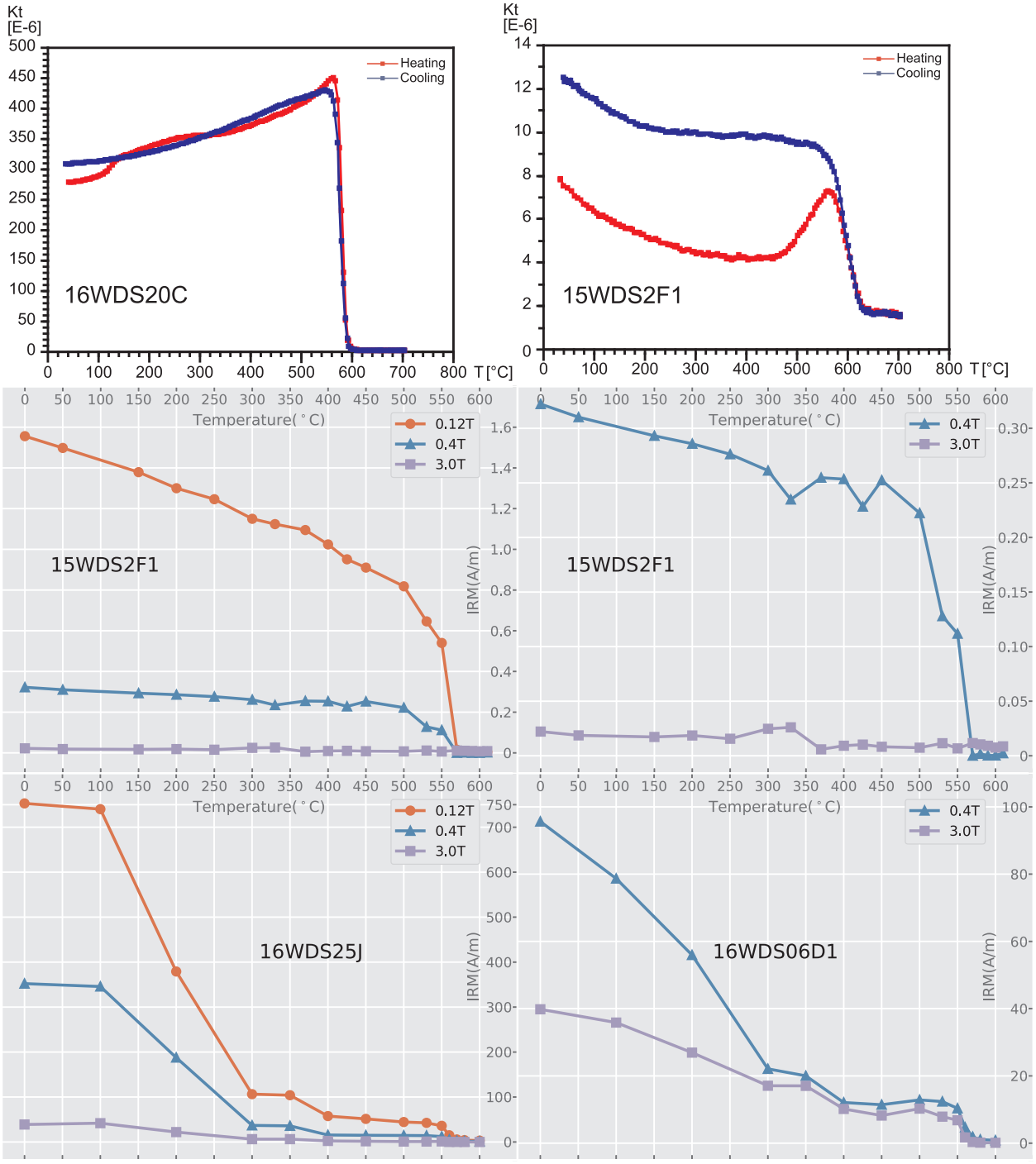


Fig. 3. Results of thermal susceptibility experiments and thermal demagnetisation of orthogonal 3-axis IRMs (Lowrie, 1990) for representative dyke samples. Examples of Lowrie tests on the right-hand side are shown without the dominant 0.12 T component.

I = −65.7°, Thébault et al. 2015, see Fig. 6a). We interpret this component as viscous remanent magnetisation acquired recently. The demagnetisation behaviour, together with rock magnetic results, suggest that the CL component is probably carried by maghemite or pyrrhotite.

The CH component, isolated from ten dykes, has unblocking temperatures (530–590 °C) typical of low-titanium titanomagnetite or pure magnetite. We interpret the CH component to be the ChRM of the Boonadgin dyke swarm. The CH component is dual-polarity, directed either SE shallow downward (4 sites) or NW shallow upward (6 sites; Fig. 6b). For simplicity, we hereafter arbitrarily refer to the SE shallow downward direction as "normal" and to its antipodal direction as "reverse" (Fig. 6). The reversal test of McFadden and McElhinny (1990) is positive with classification 'C' ($\gamma = 11.0°$, $\gamma c = 18.8°$). The dual-polarity remanence indicates that the duration of dyke swarm emplacement was sufficiently long for the geomagnetic field to reverse its polarity and therefore also for magnetic secular variation to be averaged out. Based on the results of the rock magnetic experiments, we performed AF demagnetisation up to 60 mT prior to thermal treatment for selected samples. The directions isolated with combined AF and thermal demagnetisation are identical to those isolated by exclusive thermal treatment (Fig. 5), which led us to conclude that we successfully isolated the ChRMs carried by SD/PSD (titano)magnetite.

At three sites (WDS09, 15WDS02 and 16WDS24), the host rocks were sampled for baked contact tests. Tests at two of the sites gave

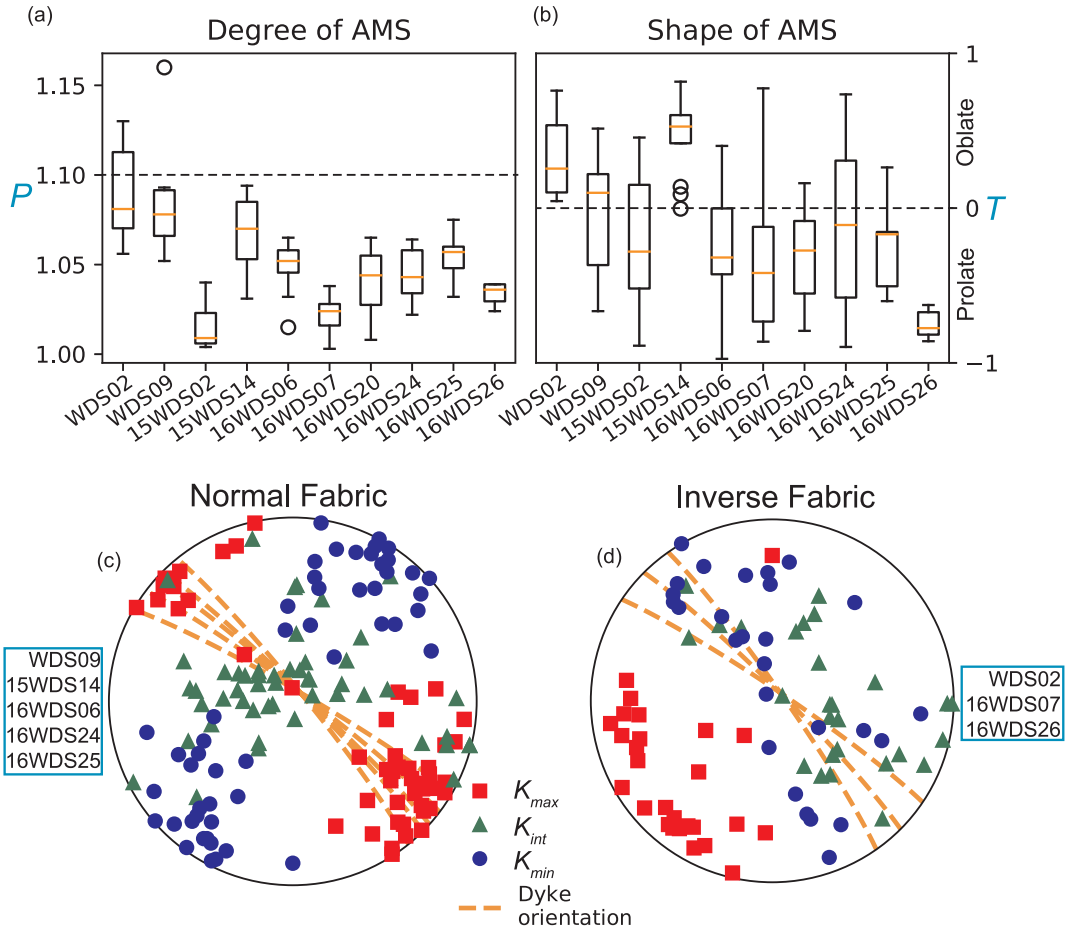


Fig. 4. Box-and-whisker plot showing (a) degree of AMS for all the sites; (b) shape of AMS for all the sites; equal-area stereonets showing principal directions of the AMS fabric for (c) all sites with normal fabric and (d) all sites with inverse fabric.

inconclusive results due to unstable magnetisations and/or randomly oriented remanence directions in the host rock. At host rock site 16WDS24, we obtained eight samples within one dyke width (10 m) away from the contact, which is the typical width of baked zones (Buchan, 2007), and seventeen samples from the hybrid and unbaked zones. Five of the eight samples from the baked zone revealed high temperature remanence components with a mean direction of D = 323.0°, I = −20.6° (α_{95} = 16.2°, k = 17.88), which is similar to the mean direction yielded by the dyke (Fig. 7b). Six of the 17 samples from outside the baked zone yielded high temperature remanence components with a mean direction of D = 4.2°, I = −43.9° (α_{95} = 11.7°, k = 33.51), which is clearly different from that of the baked zone (Fig. 7c). We note that this direction is close to that of the GAD field (Fig. 7c), but the possibility that it is a primary ancient direction of the host rock cannot be ruled out. Another possible explanation for this observation could be that the intrusion and subsequent heating of the dyke led to a mineralogical change in the baked zone of the host rock, which made it more resistant to a viscous reset of the magnetic signal. In addition, the directions for viscous overprint should be close to that of the present-day Earth magnetic field rather than that of the GAD field. The unbaked direction being closer to the GAD direction makes it less likely to be a viscous overprint (Fig. 7). Sample 16WDS24-9, located 10.5 m from the contact, shows hybrid characteristics suggestive of partial remagnetisation, with its low temperature component (< 450 °C) yielding the dyke direction and the high temperature component (500–580 °C) yielding the unbaked direction. Based on these observations, we are inclined to interpret the baked contact test as positive. However, we acknowledge that the unbaked direction is yet to be proved older than the dyke direction, which remains an important caveat to our baked contact test.

Based on the positive baked contact test, we interpret the CH component to be of primary origin. The following points also support our interpretation: (i) a positive reversal test; (ii) the dissimilarity between the direction of CH and published younger palaeomagnetic directions from the region (Fig. 6c); (iii) the fact that the nearby ca. 2.4 Ga Widgiemooltha dyke swarm preserved primary magnetisations (Smirnov et al., 2013) indicating an absence of pervasive overprinting events in the region; (iv) an unblocking temperature generally between 530 °C and 590 °C that makes the ChRM unlikely to represent a thermal overprint. After inverting the “reverse” polarity directions, the overall mean CH direction is D = 142.5°, I = 13.2° (A_{95} = 8°, k = 37.37). The corresponding pole is located at 46.8°S, 234.9°E (dp, dm = 4.2°, 8.2° and A95 = 5.9°).

5. Discussion

Palaeoproterozoic positions of the Yilgarn Craton in palaeogeographic reconstructions have been controversial due to a lack of well-constrained palaeomagnetic poles of 1900–1800 Ma antiquity. Previous reconstructions (e.g., Belica et al., 2014; Klein et al., 2016; Meert et al., 2011) incorporated either the tentative ca. 1.82 Ga Plum Tree Volcanics palaeopole from the North Australia Craton (NAC; Idnurm and Giddings, 1988), or the 1900–1800 Ma Frere Formation palaeopole (Williams et al., 2004), which has large uncertainty in the age of the magnetisation. The precisely-dated and palaeomagnetically well-defined pole from this study bridges the 1900–1800 Ma gap in the Precambrian Australian palaeomagnetic database and can be used to improve the palaeogeographic evolution for this time. Here we establish a

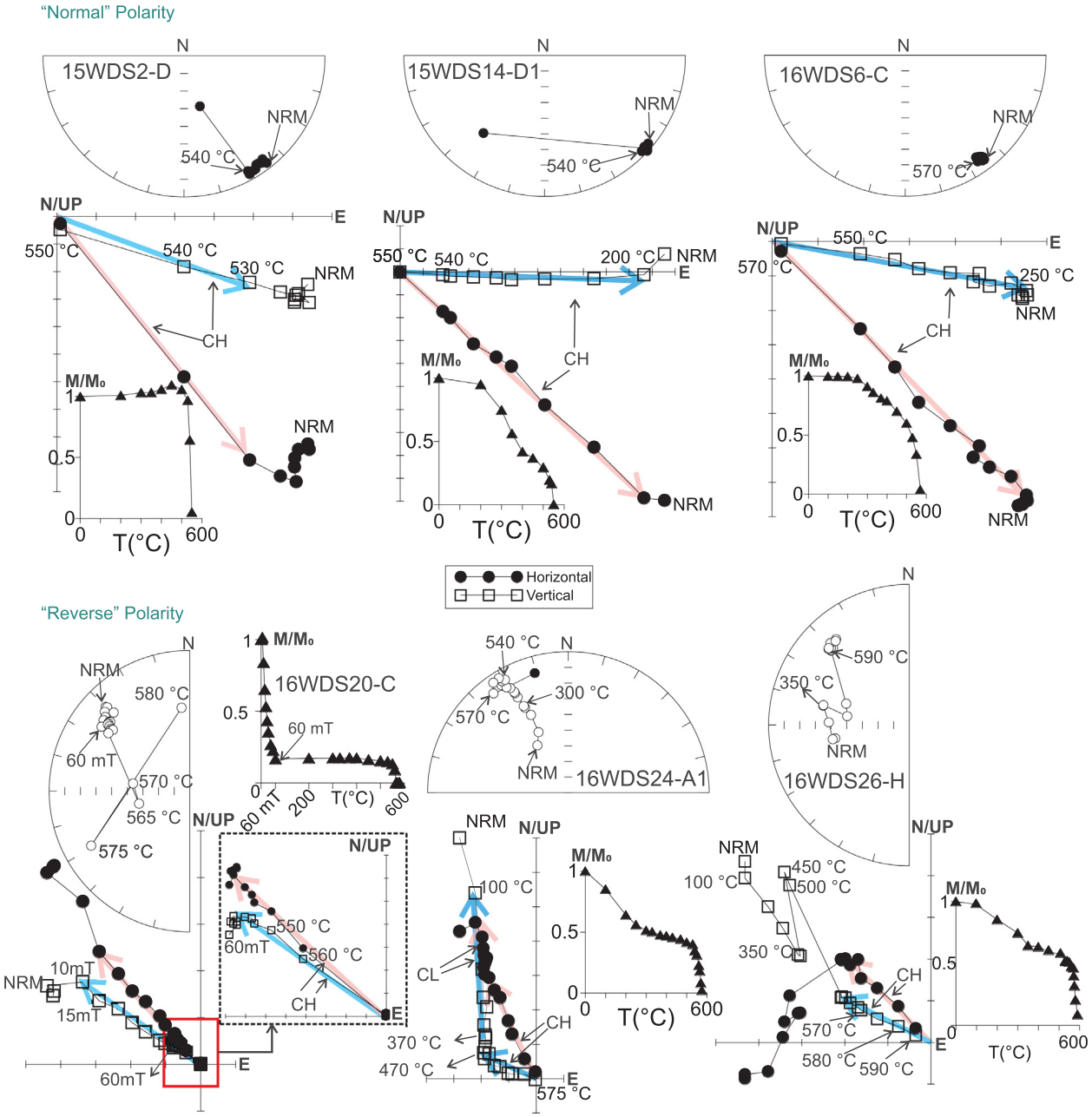


Fig. 5. Representative demagnetisation experiments shown on Zijderveld diagrams (Zijderveld, 1967). Open/closed symbols represent magnetisation vectors projected on the vertical/horizontal plane. In stereonets (equal-area projection), open/filled symbols indicate upper/lower hemisphere directions. Linear plots with filled triangles showing the normalized intensity decay curves.

global reconstruction ca. 1.89 Ga and, based on this, discuss the possible connection of the WAC with the South Indian Block (SIB).

Palaeoproterozoic palaeogeography is controversial largely due to the lack of high-quality palaeomagnetic poles. In particular, only one ~1900–1870 Ma well-dated and reliable palaeopole from the Molson dykes of the Superior Craton (Halls and Heaman, 2000; Zhai et al., 1994) had been reported until recently. However, in the last eight years several palaeomagnetic studies of coeval rocks on various continents have been published (Table 2). The palaeogeographic reconstructions proposed in these publications, however, are significantly different from each other. Nonetheless, as palaeolatitudes and azimuthal orientations of ancient continents in reconstructions of these workers are broadly similar, slight differences could be attributable to magnetic polarity ambiguity and longitudinal uncertainty. Here we propose the most up-to-date ca. 1.89 Ga palaeogeography for all cratons for which high-quality palaeopoles are available (Table 2), while also incorporating the timing of Palaeoproterozoic orogens (Table 3).Table 4.

Ca. 1.89 Ga, Laurentia and Baltica were not yet assembled (Evans and Mitchell, 2011). Therefore, poles from their building blocks should be treated separately. The location of the Superior Craton at moderate palaeolatitudes is constrained by an recalculated pole of the 1877 +7/−4 Ma Molson dykes (Evans and Halls, 2010), which is based on 34 sites from previous studies and proved primary by positive baked-contact tests (Halls and Heaman, 2000; Zhai et al., 1994). Kilian et al. (2016) reported a pole for the 1899 ± 5 Ma Sourdough dykes from the Wyoming Craton with positive baked-contact tests supporting the primary origin of the magnetisation. This pole places the Wyoming Craton at approximately the same latitudes as the Superior Craton. Here we adopt Kilian et al.'s (2016) favoured position by placing the Wyoming and Superior cratons ~60° apart in arc length, which does not require

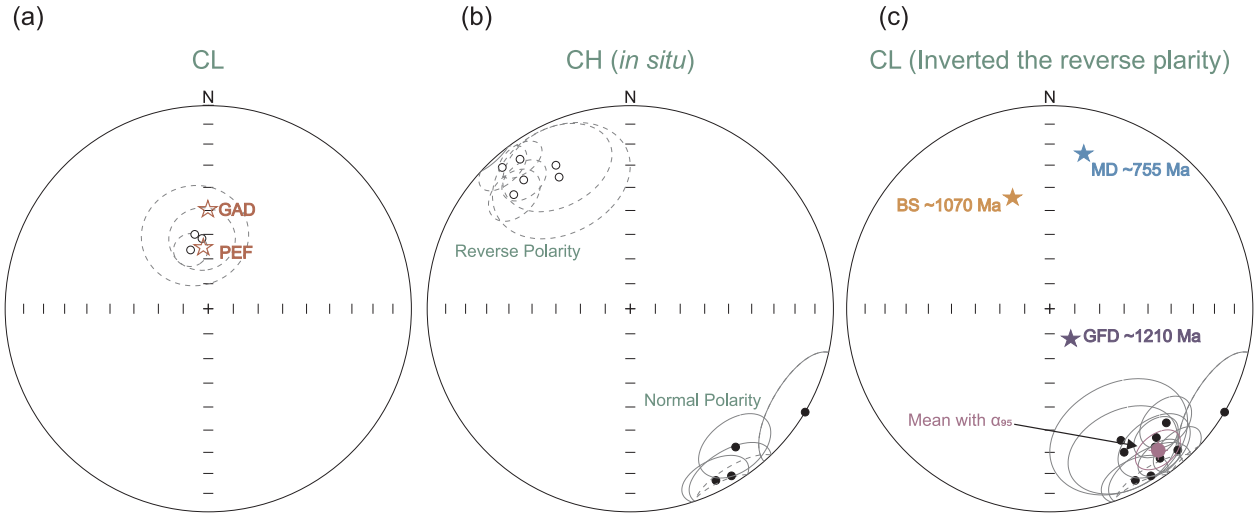


Fig. 6. Stereonets (equal-area projection) of: (a) site mean directions of CL component with stars showing the direction of the present-day Earth magnetic field (PEF) and the expected direction from geocentric axial dipole (GAD); (b) site mean directions of bipolar CH component (Table 1); (c) site mean direction of CH component with NW upward directions inverted. Filled stars represent published younger palaeomagnetic directions in the region (Table 1). GFD = 1210 Ma Gnowangerup-Fraser Dykes (Pisarevsky et al., 2014b), BS = 1070 Ma Bangemall Sills (Wingate et al., 2004, 2002), MD = 755 Ma Mundine Well Dykes (Wingate and Giddings, 2000). Conventions follow those in Fig. 5.

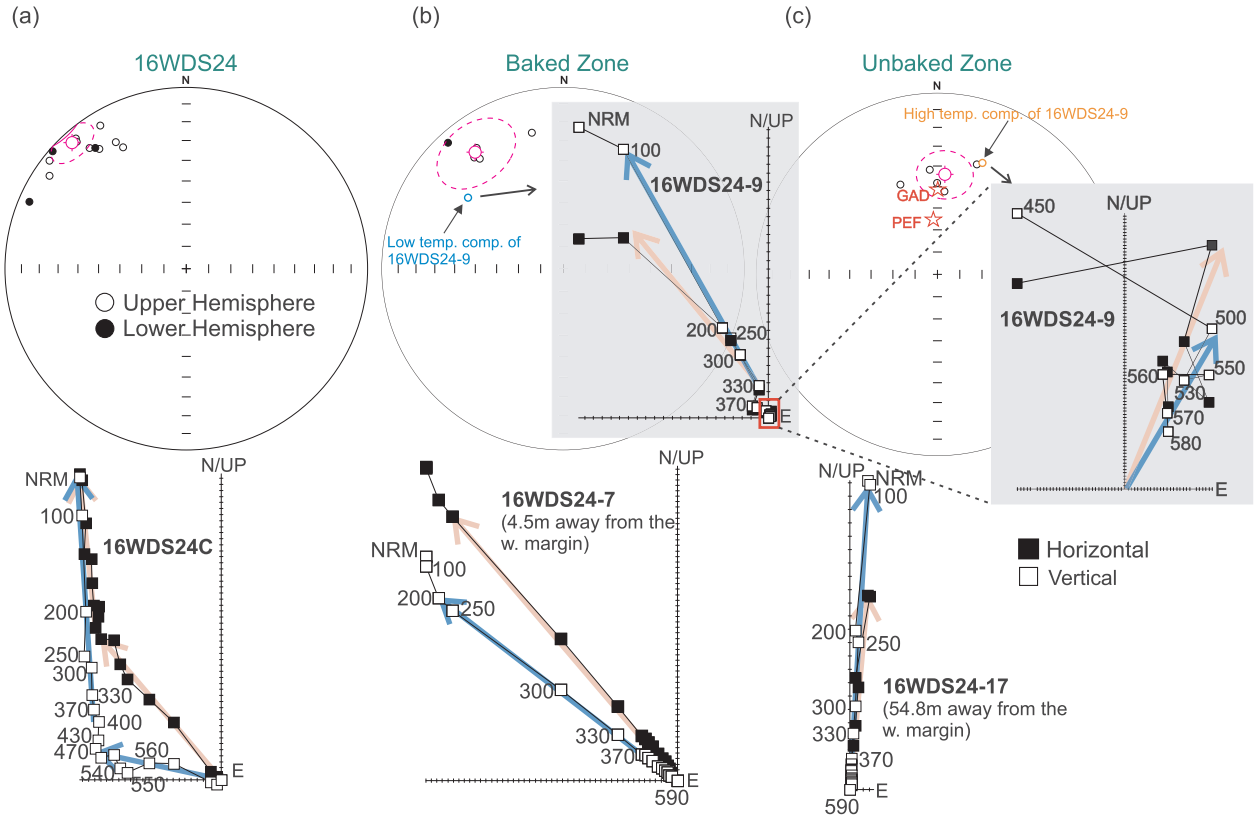


Fig. 7. Results of a backed contact test. The stereonets (equal-area projection) are of: (a) sample mean directions of site 16WDS24 (Table 1); (b) sample mean directions of the host gneiss samples collected within 10.1 m (typical baked zone) from the western margin of 16WDS24 (Table 1); (c) sample mean directions of the host gneiss samples collected farther than 10.1 m from the western margin of 16WDS24 (Table 1). The Zijderveld diagrams show progressive thermal demagnetisation results of representative samples from the dyke (16WDS24-C), the baked zone (16WDS24-7), the hybrid zone (16WDS24-9), and the unbaked country rock (16WDS24-17), respectively. Numbers labelled on the Zijderveld plots indicate the thermal demagnetisation steps in °C.

complex rotations of the Wyoming Craton in order to join the Superior Craton later, during the ca. 1770 Ma Big Sky Orogeny (Hanson, 2004). Buchan et al., (2016) reported a high-quality pole supported by multiple positive baked-contact tests for the 1885 Ma Ghost dykes of the Slave Craton. The Ghost palaeopole indicates that the Slave Craton occupied moderately low palaeolatitudes ca. 1.89 Ga. The Slave Craton collided with the Rae Craton along the Thelon Orogen ca 1.96 Ga (Hoffman, 1988), and then the Hearne Craton collided with the Rae Craton ca 1.9 Ga, forming the northwestern part of Laurentia (Berman et al., 2007), based on which we consider the Ghost dykes palaeopole to

Table 2
Palaeomagnetic poles used in the palaeographic reconstructions at ca 1.89 Ga (Fig. 8).

Pole	Cont./Craton	Plat. (°N)	Plong. (°E)	A_{95} (°)	Age (Ma)	Reference
Molson Dykes B + C2	Superior	28.9	218.00	3.8	1884–1873	Evans and Halls (2010); Halls and Heaman (2000); Zhai et al. (1994)
Sourdough Dykes	Wyoming	49.2	291.0	8.1	1904–1894	Kilian et al. (2016)
Ghost Dykes	Slave	2.0	254.0	6.0	1887–1884	Buchan et al. (2016)
Keuruu Dykes	Fennoscandia	45.7	230.9	5.5	1879–1859	Klein et al. (2016)
lower Akitan Group	Siberia	−30.8	98.7	3.5	1882–1874	Didenko et al. (2009)
Mashonaland Sills	Kalahari	8.0	338.0	5.0	1888–1874	Bates and Jones, (1996); Evans et al., (2002); McElhinny and Opdyke, (1964); Söderlund et al., (2010)
Dharwar + Bastar Dykes	India	37	334	5.6	1888–1882	Belica et al. (2014); Meert et al. (2011)
Boonadagin Dykes	W. Australia	−46.8	234.9	5.9	1892–1884	This study

Table 3
Palaeoproterozoic orogenies plotted in Fig. 8.

Name	Duration	Type	Reference
Snowbird	1920–1890 Ma	Collisional	Berman et al. (2007); Martel et al. (2008)
Wopmay	1950–1840 Ma	Accretionary	Bowring and Podosek (1989); Bowring and Grotzinger (1992)
Torngat	1940–1870 Ma	Collisional	Connelly (2001); Funck et al. (2000)
Trans-Hudson	1910–1810 Ma	Accretionary stage	Hoffman (1988)
Svecofennian	2000–1750 Ma	Accretionary	Bogdanova et al. (2015)
Kheis-Okwa-Magondi	2000–1850 Ma	Accretionary	Jacobs et al. (2008)
Angara	1900–1850 Ma	Accretionary	Gladkochub et al. (2006); Poller et al. (2005)
Akitkan	1900–1870 Ma	Collisional	Donskaya et al. (2009)

Table 4
Euler rotation parameters for Fig. 8.

Craton/block/terrane*	Euler Pole		Angle (°)
	(°N)	(°E)	
Superior	−13.34	−72.92	−235.05
Nain	−37.60	−2.14	−235.87
Rae	−29.48	−37.79	−242.56
Hearn	−29.48	−37.79	−242.56
Slave	28.35	132.16	−104.91
Wyoming	−11.83	−7.18	142.70
Fennoscandia	−64.78	49.87	−175.78
Siberia	−49.41	53.82	256.44
Kalahari	−13.78	−92.07	−105.47
S. India	43.31	−88.68	68.98
W. Australia	−40.29	161.40	−63.59

* Rotation relative to absolute framework.

represent the Slave, Rae, and Hearne cratons ca. 1.89 Ga.

The collision of the Archaean Kola and Karelian cratons during the 1940–1860 Ma Lapland-Kola Orogeny, followed by Svecofennian accretionary growth (e.g. Bogdanova et al., 2015; Lahtinen et al., 2008), formed Fennoscandia, which in turn collided with the Sarmatia/Volgo-Uralia Craton to assemble Baltica between 1800 Ma and 1700 Ma. A newly available key pole supported by baked contact tests from the ca. 1.87 Keuruu dykes (Klein et al., 2016) places Fennoscandia at low latitudes. The longitudinal position of Fennoscandia is chosen in a way so it can subsequently join Laurentia to form the so-called NENA (North Europe-North America) connection (Gower et al., 1990), which is suggested to have lasted from 1.8 Ga to 1.2 Ga (Evans and Pisarevsky 2008; Pisarevsky and Bylund, 2010; Salminen et al., 2014).

For Siberia, we use the pole from the lower Akitan Group, which is dated at 1878 ± 4 Ma and supported by positive intra-formational conglomerate test and by fold tests (Didenko et al., 2009). This pole suggests Siberia was equatorial. In our reconstruction, we show only the northwestern part of the Siberian Craton (the Angara-Anabar block), because although the Aldan block probably collided with the Angara-Anabar block along the Akitkan suture by ca. 1870 Ma (e.g., Pisarevsky et al., 2008 and references therein), palaeomagnetic studies suggest some differential rotation within the Aldan block up to ca. 1720 Ma (Pavlov et al., 2008). Evans and Mitchell (2011) proposed that Siberia could be tightly connected to Laurentia and Baltica, forming the core of Nuna (but also see Pisarevsky et al., 2008 for alternative reconstructions). Ernst et al. (2016) proposed a similar tight connection from ~1.9 Ga to ~0.7 Ga between southern Siberia and north Laurentia based on matching coeval magmatic events. Here we adopt this idea by placing the southern tip of Siberia close to the northern part of the Slave Craton, which is permitted by palaeomagnetic data.

A recalculated pole from the Mashonaland sills (Evans et al., 2002 and references therein) dated at 1888 ± 1 Ma (Söderlund et al., 2010) place Zimbabwe at moderate latitudes. Two roughly coeval poles are available for the Kaapvaal Craton. The Black Hills dyke pole (Lubnina et al., 2010) was used to constrain the location of Kaapvaal ca. 1.88 Ga (e.g. Belica et al., 2014). However, more detailed geochronologic investigations revealed a refined age of 1844.4 ± 2.6 Ma (Olsson et al., 2016) for the site where Lubnina et al., (2010) determined the magnetisation age, rendering the corresponding pole too young for our 1.89 Ga reconstruction. The 1875 ± 4 Ma post-Waterberg dolerite palaeopole is not supported by field tests (Hanson, 2004; Söderlund et al., 2010). Consequently, although comparing the post-Waterberg and Mashonaland poles Hanson et al., (2011) suggested a > 2000 km shift between Kaapvaal and Zimbabwe cratons at ca. 1890–1870 Ma, we prefer a conservative approach and adopt the model where Zimbabwe and Kaapvaal already collided between 2.0 and 1.9 Ga along the Limpopo Belt (Söderlund et al., 2010) to form proto-Kalahari. In our reconstruction, we use only the Mashonaland pole to constrain a position of proto-Kalahari at ca. 1.89 Ga. Jacobs et al. (2008) suggested that the western margin of proto-Kalahari experienced long-lasting accretionary events from 2000 Ma to 1850 Ma, while a passive margin environment characterizes its eastern border. We therefore keep some space between proto-Kalahari and other cratons.

Although the assembly of Australia in Precambrian time is still debated, most authors consider its incorporation by or shortly after ~1.8 Ga (Betts et al., 2016; Cawood and Korsch, 2008; Li and Evans, 2011; Li, 2000; Myers et al., 1996). Therefore, our new pole represents the WAC alone, and not Australia at large. A SWEAT-like reconstruction of Australia and Laurentia has been suggested to be possible at ca. 1600 Ma within the Nuna supercontinent (Betts et al., 2008; Goodge et al., 2008; Hamilton and Buchan, 2010; Payne et al., 2009; Pisarevsky et al., 2014a,b). Our reconstruction thus places the WAC at a considerable distance from the northwestern Laurentian building blocks

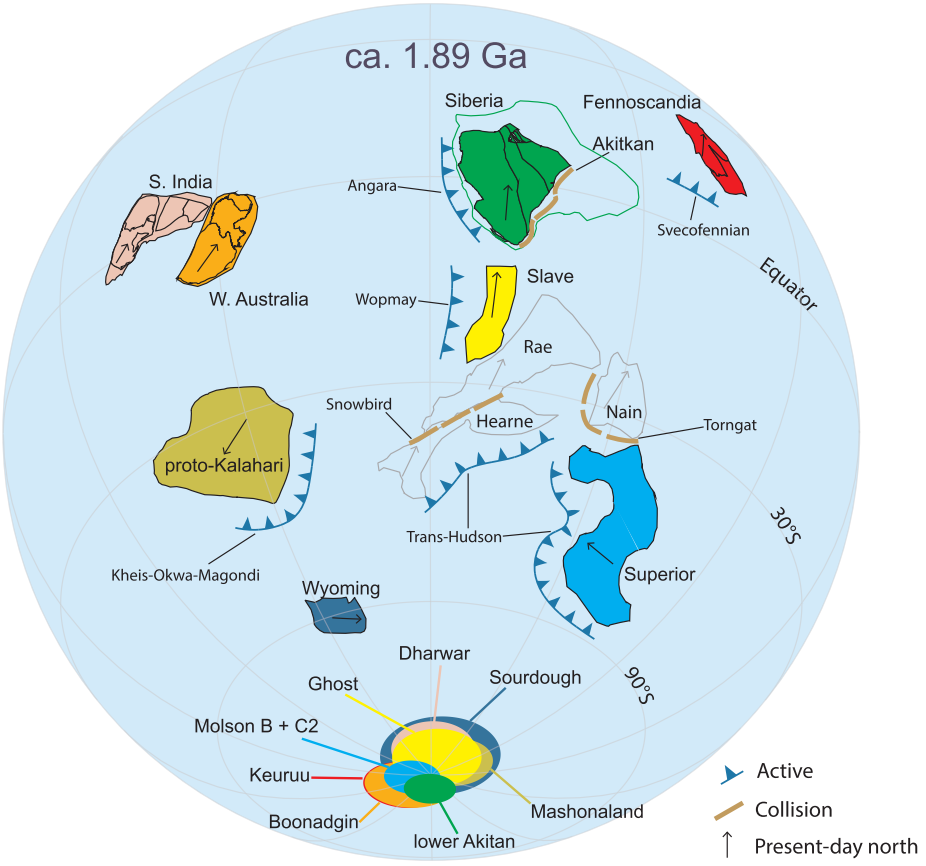


Fig. 8. Palaeogeographic reconstruction for ca. 1.89 Ga based on palaeopoles listed in Table 2. The poles are colour-coded according to the colours of the cratons. The positions of Rae, Hearne, and Nain are not palaeomagnetically constrained; their proximity to the Slave craton is nonetheless established by either active or eventual suturing (Hoffman, 1988).

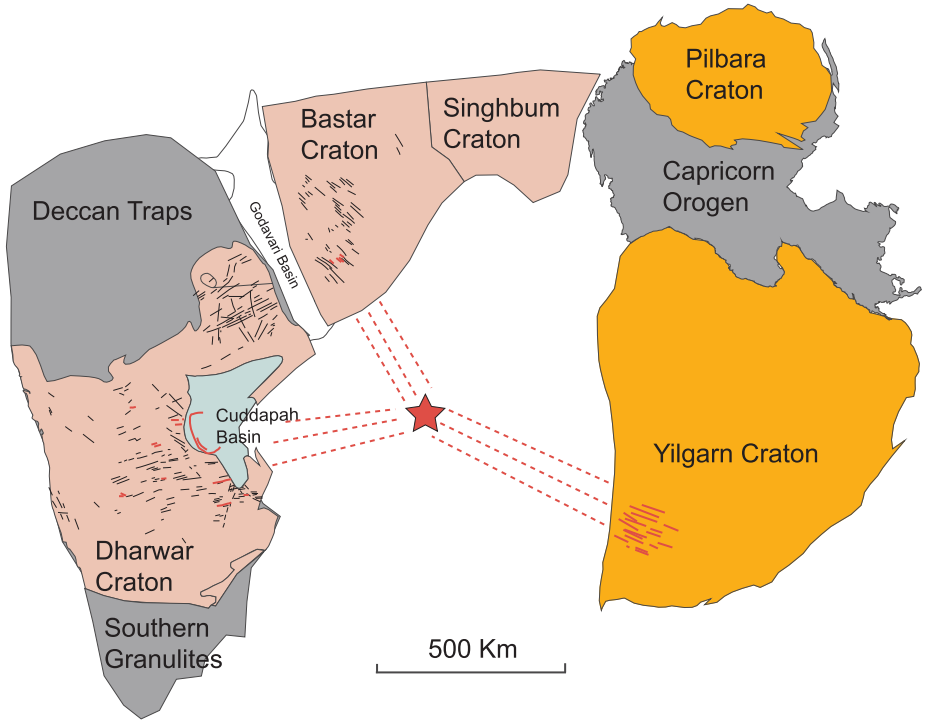


Fig. 9. A possible configuration of the WAC and SIB at ca. 1.89 Ga reconstructed in present day WAC coordinates. The red dykes in SIB have been dated at 1894–1879 Ma (Belica et al., 2014; French et al., 2008; Halls et al., 2007). Red star denotes possible location of a mantle plume centre.

(Fig. 8) so that they can later form a SWEAT-like connection, after the subsequent assembly of the North and South Australian cratons with the WAC.

Mohanty (2012, 2010) suggested a possible connection between SIB and the WAC in Late Palaeoproterozoic mainly because of a similarity of tectonic histories of the Central India Tectonic Zone and the Capricorn Orogen of the WAC. Stark et al. (this issue) discussed the possibility of the Boonadgin dyke swarm being part of the Bastar-Cuddapah LIP of SIB. Our new palaeopole supports this idea. The new Boonadgin dyke pole from this study places the WAC in proximity of the palaeoequator. Meanwhile, the palaeopole from the ca. 1.88 Ga Dharwar and Bastar dykes of SIB, which is supported by a positive baked-contact test, places India at a similarly low palaeolatitude (Belica et al., 2014). If the two continents were connected at that time, the western margin of the WAC can be reconstructed in the vicinity of the eastern margin of SIB (Fig. 9). We therefore propose that the northern WAC (Pilbara) and north-eastern India (Singhbhum) were connected or close to each other at ca. 1.9 Ga. The presence of layered intrusions and dykes of varying orientations in SIB may suggest proximity to the mantle plume centre. On the other hand, the horizontal-to-subhorizontal magma flows indicated by AMS data imply that the WAC was relatively further away from the plume centre (Ernst and Baragar, 1992). The speculated distances of South India and the WAC relative to the plume centre is consistent with our reconstruction (Fig. 9). We note that this reconstruction does not support the relation between the Central Indian and the Capricorn orogens as suggested by Mohanty (2012, 2010), and it allows additional continental block(s) to be between the Yilgarn Craton and SIB. In summary, our reconstruction in Fig. 8 does not support the “early” (Palaeoproterozoic) assembly of Nuna, but rather supports largely independently drifting cratons separated by oceans at ca. 1.89 Ga.

6. Conclusion

We obtained a palaeomagnetic key pole from the 1888 ± 9 Ma Boonadgin dyke swarm in the Yilgarn Craton, Western Australia, located at 47°S, 235°E, A_{95} = 6°. This palaeopole is proven to be primary by a positive baked-contact test. Based on matching geometries of contemporaneous mafic dykes and comparing palaepoles, we propose that the WAC (Pilbara) and SIB (Singhbhum) were close to each other ca. 1.89 Ga. Using available high-quality poles, we provide a ca. 1.89 Ga palaeogeographic reconstruction in which the WAC was positioned at a significant distance from the northwestern building blocks of Laurentia in a way which allows the subsequent amalgamation to form the proto-SWEAT connection.

Acknowledgments

This is contribution 1123 from the ARC Centre of Excellence for Core to Crust Fluid Systems (http://www.ccfs.mq.edu.au). This work was funded by ARC Centre for Core to Crust Fluid Systems CoE Grant 663 (CE110001017, Flagship Program 5) and ARC Laureate Fellowship Grant (FL150100133) to Zheng-Xiang Li. We thank Jie Peng for his support with thermomagnetic analyses. We also thank Jiangyu Li for his assistance with sample collection in the field. Constructive comments from Natalia V. Lubnina, an anonymous reviewer and guest editor Richard E. Ernst greatly improved the manuscript and are gratefully acknowledged.

References

Bates, M.P., Jones, D.L., 1996. A palaeomagnetic investigation of the Mashonaland dolerites, north-east Zimbabwe. Geophys. J. Int. 126, 513–524. http://dx.doi.org/10.1111/j.1365-246X.1996.tb05307.x.

Belica, M.E., Piispa, E.J., Meert, J.G., Pesonen, L.J., Plado, J., Pandit, M.K., Kamenov, G.D., Celestino, M., 2014. Paleoproterozoic mafic dyke swarms from the Dharwar craton; paleomagnetic poles for India from 2.37 to 1.88Ga and rethinking the Columbia supercontinent. Precambr. Res. 244, 100–122. http://dx.doi.org/10.1016/j.precamres.2013.12.005.

Berman, R.G., Davis, W.J., Pehrsson, S., 2007. Collisional Snowbird tectonic zone resurrected: Growth of Laurentia during the 1.9 Ga accretionary phase of the Hudsonian orogeny. Geology 35, 911. http://dx.doi.org/10.1130/G23771A.1.

Betts, P.G., Armit, R.J., Stewart, J., Aitken, A.R.A., Ailleres, L., Donchak, P., Hutton, L., Withnall, I., Giles, D., 2016. Australia and Nuna. Geol. Soc. London, Spec. Publ. 424, 47–81. http://dx.doi.org/10.1144/SP424.2.

Betts, P.G., Giles, D., Schaefer, B.F., 2008. Comparing 1800–1600 Ma accretionary and basin processes in Australia and Laurentia: Possible geographic connections in Columbia. Precambr. Res. 166, 81–92. http://dx.doi.org/10.1016/j.precamres.2007.03.007.

Bleeker, W., Ernst, R., 2006. Short-lived mantle generated magmatic events and their dyke swarms: The key unlocking Earth’s palaeogeographic record back to 2.6 Ga. In: Symposium Volume for Fifth International Dyke Conference, July–August 2005. Balkema, Rotterdam, pp. 3–26.

Bogdanova, S., Gorbatschev, R., Skridlaite, G., Soesoo, A., Taran, L., Kurlovich, D., 2015. Trans-Baltic Palaeoproterozoic correlations towards the reconstruction of supercontinent Columbia/Nuna. Precambr. Res. 259, 5–33. http://dx.doi.org/10.1016/j.precamres.2014.11.023.

Bowring, S.A., Grotzinger, J.P., 1992. Implications of new chronostratigraphy for tectonic evolution of Wopmay Orogen, Northwest Canadian Shield. Am. J. Sci. 292, 1–20. http://dx.doi.org/10.2475/ajs.292.1.1.

Bowring, S.A., Podosek, A.F., 1989. Nd isotopic evidence from Wopmay Orogen for 2.0–2.4 Ga crust in western North America. Earth Planet. Sci. Lett. 94, 217–230. http://dx.doi.org/10.1016/0012-821X(89)90141-6.

Boyd, D.M., Tucker, D.H., 1990. Australian Magnetic Dykes. In: Parker, A.J., Rickwood, P.C., Tucker, D.H. (Eds.), Mafic Dykes and Emplacement Mechanisms. CRC Press, pp. 391–399.

Buchan, K.L., 2007. Baked contact test. In: Gibbins, D., Herrero-Bervera, E. (Eds.), Encyclopedia of Geomagnetism and Paleomagnetism. Springer, Dordrecht, Netherlands, pp. 35–39.

Buchan, K.L., Mitchell, R.N., Bleeker, W., Hamilton, M.A., LeCheminant, A.N., 2016. Paleomagnetism of ca. 2.13–2.11 Ga Indin and ca. 1.885 Ga Ghost dyke swarms of the Slave craton: implications for the Slave craton APW path and relative drift of Slave, Superior and Siberian cratons in the Paleoproterozoic. Precambr. Res. 275, 151–175. http://dx.doi.org/10.1016/j.precamres.2016.01.012.

Canon-Tapia, E., 2004. Anisotropy of magnetic susceptibility of lava flows and dykes: a historical account. Geol. Soc. London, Spec. Publ. 238, 205–225. http://dx.doi.org/10.1144/GSL.SP.2004.238.01.14.

Cawood, P.A., Korsch, R.J., 2008. Assembling Australia: Proterozoic building of a continent. Precambr. Res. 166, 1–35. http://dx.doi.org/10.1016/j.precamres.2008.08.006.

Chadima, M., Cajz, V., Týcová, P., 2009. On the interpretation of normal and inverse magnetic fabric in dikes: examples from the Eger Graben, NW Bohemian Massif. Tectonophysics 466, 47–63. http://dx.doi.org/10.1016/j.tecto.2008.09.005.

Chadima, M., Hrouda, F., 2006. Remasoft 3.0 a user-friendly paleomagnetic data browser and analyzer. Trav. Géophysiques 27, 20–21.

Chadima, M., Jelínek, V. 2008. Anisoft 4.2.–Anisotropy data browser. In: Paleo, Rock and Environmental Magnetism, 11th Castle Meeting, Contribution to Geophysics and Geodesy, Special Issue. Geophysical Institute of the Slovak Academy of Sciences, Bojnice Castle, Slovac Republic.

Chen, S.F., Riganti, A., Wyche, S., Greenfield, J.E., Nelson, D.R., 2003. Lithostratigraphy and tectonic evolution of contrasting greenstone successions in the central Yilgarn Craton, Western Australia. Precambr. Res. 127, 249–266. http://dx.doi.org/10.1016/S0301-9268(03)00190-6.

Connelly, J.N., 2001. Constraining the Timing of Metamorphism: U-Pb and Sm-Nd Ages from a Transect across the Northern Torngat Orogen, Labrador, Canada. J. Geol. 109, 57–77. http://dx.doi.org/10.1086/317965.

Didenko, A.N., Vodovozov, V.Y., Pisarevsky, S.A., Gladkochub, D.P., Donskaya, T.V., Mazukabzov, A.M., Stanevich, A.M., Bibikova, E.V., Kirnozova, T.I., 2009. Palaeomagnetism and U-Pb dates of the Palaeoproterozoic Akitkan Group (South Siberia) and implications for pre-Neoproterozoic tectonics. Geol. Soc. London. Spec. Publ. 323, 145–163. http://dx.doi.org/10.1144/SP323.7.

Donskaya, T.V., Gladkochub, D.P., Pisarevsky, S.A., Poller, U., Mazukabzov, A.M., Bayanova, T.B., 2009. Discovery of Archaean crust within the Akitkan orogenic belt of the Siberian craton: New insight into its architecture and history. Precambr. Res. 170, 61–72. http://dx.doi.org/10.1016/j.precamres.2008.12.003.

Dunlop, D.J., Ozdemir, O., 1997. Rock Magnetism: Fundamentals and Frontiers. Cambridge University Press, Cambridge doi:10.1017/CBO9780511612794.

Ernst, R., Srivastava, R., Bleeker, W., Hamilton, M., 2010. Precambrian Large Igneous Provinces (LIPs) and their dyke swarms: new insights from high-precision geochronology integrated with paleomagnetism and geochemistry. Precambr. Res. 183. http://dx.doi.org/10.1016/j.precamres.2010.09.001.

Ernst, R.E., Baragar, W.R.A., 1992. Evidence from magnetic fabric for the flow pattern of magma in the Mackenzie giant radiating dyke swarm. Nature 356, 511–513. http://dx.doi.org/10.1038/356511a0.

Ernst, R.E., Hamilton, M.A., Soderlund, U., Hanes, J.A., Gladkochub, D.P., Okrugin, A.V., Kolotilina, T., Mekhonoshin, A.S., Bleeker, W., LeCheminant, A.N., Buchan, K.L., Chamberlain, K.R., Didenko, A.N., 2016. Long-lived connection between southern Siberia and northern Laurentia in the Proterozoic. Nat. Geosci 9, 464–469. http://dx.doi.org/10.1038/ngeo2700/rhttp://www.nature.com/ngeo/journal/v9/n6/abs/ngeo2700.html#supplementary-information.

Evans, D.A.D., Pisarevsky, S.A., 2008. Plate tectonics on early Earth? Weighing the paleomagnetic evidence Plate tectonics on early Earth? Weighing the paleomagnetic evidence. Society 2440, 249–263. http://dx.doi.org/10.1130/2008.2440(12).

Evans, D.A.D., Beukes, N.J., Kirschvink, J.L., 2002. Paleomagnetism of a lateritic paleoweathering horizon and overlying Paleoproterozoic red beds from South Africa: Implications for the Kaapvaal apparent polar wander path and a confirmation of atmospheric oxygen enrichment. J. Geophys. Res. Solid Earth 107. http://dx.doi.org/10.1029/2001JB000432.

Evans, D.A.D., Halls, H.C., 2010. Restoring Proterozoic deformation within the Superior craton. Precambr. Res. 183, 474–489. http://dx.doi.org/10.1016/j.precamres.2010.02.007.

Evans, D.A.D., Li, Z.X., Murphy, J.B., 2016. Four-dimensional context of Earth's supercontinents. Geol. Soc. London, Spec. Publ. 424, 1–14. http://dx.doi.org/10.1144/SP424.12.

Evans, D.A.D., Mitchell, R.N., 2011. Assembly and breakup of the core of Paleoproterozoic-Mesoproterozoic supercontinent Nuna. Geology 39, 443–446. http://dx.doi.org/10.1130/G31654.1.

Evans, M.E., 1968. Magnetization of dikes: a study of the paleomagnetism of the Widgiemooltha Dike Suite, western Australia. J. Geophys. Res. 73, 3261–3270. http://dx.doi.org/10.1029/JB073i010p03261.

Fisher, R., 1953. Dispersion on a sphere. In: Proceedings of the Royal Society of London A: Mathematical, Physical and Engineering Sciences. The Royal Society, pp. 295–305.

French, J.E., Heaman, L.M., Chacko, T., Srivastava, R.K., 2008. 1891–1883Ma Southern Bastar-Cuddapah mafic igneous events, India: a newly recognized large igneous province. Precambr. Res. 160, 308–322. http://dx.doi.org/10.1016/j.precamres.2007.08.005.

Funck, T., Louden, K.E., Wardle, R.J., Hall, J., Hobro, J.W., Salisbury, M.H., Muzzatti, A.M., 2000. Three-dimensional structure of the Torngat Orogen (NE Canada) from active seismic tomography. J. Geophys. Res. Solid Earth 105, 23403–23420. http://dx.doi.org/10.1029/2000JB900228.

Giddings, J.W., 1976. Precambrian palaeomagnetism in Australia I: Basic dykes and volcanics from the Yilgarn Block. Tectonophysics 30, 91–108. http://dx.doi.org/10.1016/0040-1951(76)90138-4.

Gladkochub, D., Pisarevsky, S., Donskaya, T., Natapov, L., Mazukabzov, A., Stanevich, A., Sklyarov, E., 2006. The Siberian Craton and its evolution in terms of the Rodinia hypothesis. Episodes 29, 169–174.

Goodge, J.W., Vervoort, J.D., Fanning, C.M., Brecke, D.M., Farmer, G.L., Williams, I.S., Myrow, P.M., DePaolo, D.J., 2008. A positive test of east Antarctica-Laurentia juxtaposition within the Rodinia supercontinent. Science (80-.) 321, 235–240. http://dx.doi.org/10.1126/science.1159189.

Gower, C.F., Ryan, A.B., Rivers, T., 1990. Mid-Proterozoic Laurentia-Baltica: an overview of its geological evolution and a summary of the contributions made by this vlolume. Geol. Assoc. Canada Spec. Pap. 1–20.

Halls, H.C., Heaman, L.M., 2000. The paleomagnetic significance of new U-Pb age data from the Molson dyke swarm, Cauchon Lake area. Manitoba. Can. J. Earth Sci. 37, 957–966. http://dx.doi.org/10.1139/e00-010.

Halls, H.C., Kumar, A., Srinivasan, R., Hamilton, M., a., 2007. Paleomagnetism and U-Pb geochronology of easterly trending dykes in the Dharwar craton, India: feldspar clouding, radiating dyke swarms and the position of India at 2.37 Ga. Precambr. Res. 155, 47–68. http://dx.doi.org/10.1016/j.precamres.2007.01.007.

Halls, H.C., Wingate, M.T.D., 2001. Paleomagnetic pole from the yilgarn B (YB) dykes of Western Australia: No Longer relevant to Rodinia reconstructions. Earth Planet. Sci. Lett. 187, 39–53. http://dx.doi.org/10.1016/S0012-821X(01)00279-5.

Hamilton, M.A., Buchan, K.L., 2010. U-Pb geochronology of the Western Channel Diabase, northwestern Laurentia: Implications for a large 1.59Ga magmatic province, Laurentia's APWP and paleocontinental reconstructions of Laurentia, Baltica and Gawler craton of southern Australia. Precambr. Res. 183, 463–473. http://dx.doi.org/10.1016/j.precamres.2010.06.009.

Hanson, R.E., 2004. Paleoproterozoic intraplate magmatism and basin development on the Kaapvaal Craton: Age, paleomagnetism and geochemistry of 1.93 to 1.87 Ga post-Waterberg dolerites. South African J. Geol. 107, 233–254. http://dx.doi.org/10.2113/107.1-2.233.

Hanson, R.E., Rioux, M., Gose, W.A., Blackburn, T.J., Bowring, S.A., 2011. Paleomagnetic and geochronological evidence for large-scale post – 1. 88 Ga displacement between the Zimbabwe and Kaapvaal cratons along the Limpopo belt. Geology 39, 487–490. http://dx.doi.org/10.1130/G31698.1.

Hoffman, P.F., 1988. United plates of America, the birth of a craton: early proterozoic assembly and growth of Laurentia. Annu. Rev. Earth Planet. Sci. 16, 543–603. http://dx.doi.org/10.1146/annurev.ea.16.050188.002551.

Idnurm, M., Giddings, J., 1988. Australian Precambrian polar wander: a review. Precambr. Res. 40–41, 61–88. http://dx.doi.org/10.1016/0301-9268(88)90061-7.

Jacobs, J., Pisarevsky, S., Thomas, R.J., Becker, T., 2008. The Kalahari Craton during the assembly and dispersal of Rodinia. Precambr. Res. 160, 142–158. http://dx.doi.org/10.1016/j.precamres.2007.04.022.

Jeffery, G.B., 1922. The Motion of Ellipsoidal Particles Immersed in a Viscous Fluid. Proc. R. Soc. A Math. Phys. Eng. Sci. 102, 161–179. http://dx.doi.org/10.1098/rspa.1922.0078.

Johnson, S.P., Sheppard, S., Rasmussen, B., Wingate, M.T.D., Kirkland, C.L., Muhling, J.R., Fletcher, I.R., Belousova, E.A., 2011. Two collisions, two sutures: Punctuated pre-1950Ma assembly of the West Australian Craton during the Ophthalmian and Glenburgh Orogenies. Precambr. Res. 189, 239–262. http://dx.doi.org/10.1016/j.precamres.2011.07.011.

Johnson, S.P., Thorne, A.M., Tyler, I.M., Korsch, R.J., Kennett, B.L.N., Cutten, H.N., Goodwin, J., Blay, O., Blewett, R.S., Joly, A., Dentith, M.C., Aitken, A.R.A., Holzschuh, J., Salmon, M., Reading, A., Heinson, G., Boren, G., Ross, J., Costelloe, R.D., Fomin, T., 2013. Crustal architecture of the Capricorn Orogen, Western Australia and associated metallogeny. Aust. J. Earth Sci. 60, 681–705. http://dx.doi.org/10.1080/08120099.2013.826735.

Kilian, T.M., Chamberlain, K.R., Evans, D.A.D., Bleeker, W., Cousens, B.L., 2016. Wyoming on the run—Toward final Paleoproterozoic assembly of Laurentia. Geology 44, 863–866. http://dx.doi.org/10.1130/G38042.1.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophys. J. R. Astron. Soc. 62, 699–718. http://dx.doi.org/10.1111/j.1365-246X.1980.tb02601.x.

Klein, R., Pesonen, L.J., Mänttäri, I., Heinonen, J.S., 2016. A late Paleoproterozoic key pole for the Fennoscandian Shield: A paleomagnetic study of the Keuruu diabase dykes, Central Finland. Precambr. Res. 286, 379–397. http://dx.doi.org/10.1016/j.precamres.2016.10.013.

Knight, M.D., Walker, G.P.L., 1988. Magma flow directions in dikes of the Koolau Complex, Oahu, determined from magnetic fabric studies. J. Geophys. Res. 93, 4301. http://dx.doi.org/10.1029/JB093iB05p04301.

Lahtinen, R., Garde, A.A., Melezhik, V.A., 2008. Paleoproterozoic evolution of Fennoscandia and Greenland. Episodes 31, 20–28.

Letts, S., Torsvik, T.H., Webb, S.J., Ashwal, L.D., 2011. New Palaeoproterozoic palaeomagnetic data from the Kaapvaal Craton, South Africa. Geol. Soc. London. Spec. Publ. 357, 9–26. http://dx.doi.org/10.1144/SP357.2.

Lewis, J.D. 1994. Mafic dykes in the Williams – Wandering area, Western Australia.

Li, Z.-X., Evans, D., a D., 2011. Late Neoproterozoic 40° intraplate rotation within Australia allows for a tighter-fitting and longer-lasting Rodinia. Geology 39, 39–42. http://dx.doi.org/10.1130/G31461.1.

Li, Z.X., 2000. Palaeomagnetic evidence for unification of the North and West Australian cratons by ca.1.7 Ga: New results from the Kimberley Basin of northwestern Australia. Geophys. J. Int. 142, 173–180. http://dx.doi.org/10.1046/j.1365-246X.2000.00143.x.

Lowrie, W., 1990. Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties. Geophys. Res. Lett. 17, 159–162. http://dx.doi.org/10.1029/GL017i002p00159.

Lubnina, N., Ernst, R., Klausen, M., Söderlund, U., 2010. Paleomagnetic study of NeoArchean–Paleoproterozoic dykes in the Kaapvaal Craton. Precambr. Res. 183, 523–552. http://dx.doi.org/10.1016/j.precamres.2010.05.005.

Martel, E., van Breemen, O., Berman, R.G., Pehrsson, S., 2008. Geochronology and tectonometamorphic history of the Snowbird Lake area, Northwest Territories, Canada: New insights into the architecture and significance of the Snowbird tectonic zone. Precambr. Res. 161, 201–230. http://dx.doi.org/10.1016/j.precamres.2007.07.007.

McElhinny, M.W., Opdyke, N.D., 1964. The paleomagnetism of the precambrian dolerites of Eastern Southern Rhodesia, an example of geologic correlation by rock magnetism. J. Geophys. Res. 69, 2465–2475. http://dx.doi.org/10.1029/JZ069i012p02465.

McFadden, P.L., McElhinny, M.W., 1990. Classification of the reversal test in palaeomagnetism. Geophys. J. Int. 103, 725–729. http://dx.doi.org/10.1111/j.1365-246X.1990.tb05683.x.

Meert, J.G., Pandit, M.K., Pradhan, V.R., Kamenov, G., 2011. Preliminary report on the paleomagnetism of 1.88ga dykes from the bastar and dharwar cratons, peninsular india. Gondwana Res. 20, 335–343. http://dx.doi.org/10.1016/j.gr.2011.03.005.

Mohanty, S., 2012. Spatio-temporal evolution of the Satpura Mountain Belt of India: A comparison with the Capricorn Orogen of Western Australia and implication for evolution of the supercontinent Columbia. Geosci. Front. 3, 241–267. http://dx.doi.org/10.1016/j.gsf.2011.10.005.

Mohanty, S., 2010. Tectonic evolution of the Satpura Mountain Belt : A critical evaluation and implication on supercontinent assembly. J. Asian Earth Sci. 39, 516–526. http://dx.doi.org/10.1016/j.jseaes.2010.04.025.

Myers, J.S., 1993. Precambrian History of the West Australian Craton and Adjacent Orogens. Annu. Rev. Earth Planet. Sci. 21, 453–485. http://dx.doi.org/10.1146/annurev.ea.21.050193.002321.

Myers, J.S., Shaw, R.D., Tyler, I.M., 1996. Tectonic evolution of Proterozoic Australia. Tectonics 15, 1431–1446. http://dx.doi.org/10.1029/96TC02356.

Olsson, J.R., Klausen, M.B., Hamilton, M.A., März, N., Söderlund, U., Roberts, R.J., 2016. Baddeleyite U-Pb ages and gechemistry of the 1875–1835 Ma Black Hills Dyke Swarm across north-eastern South Africa: part of a trans-Kalahari Craton back-arc setting? GFF 138, 183–202. http://dx.doi.org/10.1080/11035897.2015.1103781.

Pavlov, V., Bachtadse, V., Mikhailov, V., 2008. New Middle Cambrian and Middle Ordovician palaeomagnetic data from Siberia: Llandelian magnetostratigraphy and relative rotation between the Aldan and Anabar-Angara blocks. Earth Planet. Sci. Lett. 276 (3–4), 229–242. http://dx.doi.org/10.1016/j.epsl.2008.06.021.

Payne, J.L., Hand, M., Barovich, K.M., Reid, A., Evans, D.A.D., 2009. Correlations and reconstruction models for the 2500–1500 Ma evolution of the Mawson Continent. Geol. Soc. London. Spec. Publ. 323, 319–355. http://dx.doi.org/10.1144/SP323.16.

Pehrsson, S.J., Eglington, B.M., Evans, D.A.D., Huston, D., Reddy, S.M., 2016. Metallogeny and its link to orogenic style during the Nuna supercontinent cycle. Geol. Soc. London. Spec. Publ. 424, 83–94. http://dx.doi.org/10.1144/SP424.5.

Pidgeon, R.T., 1990. Timing of plutonism in the Proterozoic Albany Mobile Belt, southwestern Australia. Precambr. Res. 47, 157–167. http://dx.doi.org/10.1016/0301-9268(90)90036-P.

Pisarevsky, S.A., De Waele, B., Jones, S., Söderlund, U., Ernst, R.E., 2015. Paleomagnetism and U-Pb age of the 2.4Ga Erayinia mafic dykes in the south-western Yilgarn, Western Australia: Paleogeographic and geodynamic implications. Precambr. Res. 259, 222–231. http://dx.doi.org/10.1016/j.precamres.2014.05.023.

Pisarevsky, S.A., Bylund, G., 2010. Paleomagnetism of 1780–1770 Ma mafic and composite intrusions of Smaland (Sweden): implications for the mesoproterozoic supercontinent. Am. J. Sci. 310, 1168–1186. http://dx.doi.org/10.2475/09.2010.15.

Pisarevsky, S.A., Elming, S.-Å., Pesonen, L.J., Li, Z.-X., 2014a. Mesoproterozoic paleogeography: supercontinent and beyond. Precambr. Res. 244, 207–225. http://dx.doi.org/10.1016/j.precamres.2013.05.014.

Pisarevsky, S.A., Natapov, L.M., Donskaya, T.V., Gladkochub, D.P., Vernikovsky, V.A., 2008. Proterozoic Siberia: a promontory of Rodinia. Precambr. Res. 160, 66–76. http://dx.doi.org/10.1016/j.precamres.2007.04.016.

Pisarevsky, S.A., Wingate, M., Harris, L., 2003. Late Mesoproterozoic (ca 1. 2 Ga) palaeomagnetism of the Albany – Fraser orogen : no pre-Rodinia Australia – Laurentia connection. Geophys. J. Int. 155, 6–11.

Pisarevsky, S.A., Wingate, M.T.D., Li, Z.-X., Wang, X.-C., Tohver, E., Kirkland, C.L., 2014b. Age and paleomagnetism of the 1210Ma Gnowangerup-Fraser dyke swarm, Western Australia, and implications for late Mesoproterozoic paleogeography. Precambr. Res. 246, 1–15. http://dx.doi.org/10.1016/j.precamres.2014.02.011.

Poller, U., Gladkochub, D., Donskaya, T., Mazukabzov, A., Sklyarov, E., Todt, W., 2005. Multistage magmatic and metamorphic evolution in the Southern Siberian Craton: Archean and Palaeoproterozoic zircon ages revealed by SHRIMP and TIMS. Precambr. Res. 136, 353–368. http://dx.doi.org/10.1016/j.precamres.2004.12.003.

Potter, D.K., Stephenson, A., 1988. Single-domain particles in rocks and magnetic fabric analysis. Geophys. Res. Lett. 15, 1097–1100. http://dx.doi.org/10.1029/GL015i010p01097.

Salminen, J., Mertanen, S., Evans, D.A.D., Wang, Z., 2014. Paleomagnetic and geochemical studies of the Mesoproterozoic Satakunta dyke swarms, Finland, with implications for a Northern Europe – North America (NENA) connection within Nuna supercontinent. Precambr. Res. 244, 170–191. http://dx.doi.org/10.1016/j.precamres.2013.08.006.

Sheppard, S., Bodorkos, S., Johnson, S.P., Wingate, M.T.D., Kirkland, C.L. 2010. The Paleoproterozoic Capricorn Orogeny: intracontinental reworking not continent–continent collision, Geological Survey of Western Australia Report 108.

Smirnov, A.V., Evans, D.A.D., Ernst, R.E., Söderlund, U., Li, Z.-X., 2013. Trading partners: tectonic ancestry of southern Africa and western Australia, in Archean supercratons Vaalbara and Zimgarn. Precambr. Res. 224, 11–22. http://dx.doi.org/10.1016/j.precamres.2012.09.020.

Söderlund, U., Hofmann, A., Klausen, M.B., Olsson, J.R., Ernst, R.E., Persson, P.-O., 2010. Towards a complete magmatic barcode for the Zimbabwe craton: Baddeleyite U-Pb dating of regional dolerite dyke swarms and sill complexes. Precambr. Res. 183, 388–398. http://dx.doi.org/10.1016/j.precamres.2009.11.001.

Spaggiari, C., Bodorkos, S., Barquero-Molina, M., Tyler, I., Wingate, M., 2009. Interpreted bedrock geology of the south Yilgarn and central Albany-Fraser Orogen. Western Australia, Geological Survey of Western Australia.

Stark, J.C., Wang, X.-C., Denyszyn, S.W., Li, Z.-X., Rasmussen, B., Zi, J.-W., Sheppard, S., Liu, Y., 2017. Newly identified 1.89 Ga mafic dyke swarm in the Archean Yilgarn Craton, Western Australia suggests a connection with India. Precambr. Res. http://dx.doi.org/10.1016/j.precamres.2017.12.036.

Thébault, E., Finlay, C.C., Beggan, C.D., Alken, P., Aubert, J., Barrois, O., Bertrand, F., Bondar, T., Boness, A., Brocco, L., Canet, E., Chambodut, A., Chulliat, A., Coïsson, P., Civet, F., Du, A., Fournier, A., Fratter, I., Gillet, N., Hamilton, B., Hamoudi, M., Hulot, G., Jager, T., Korte, M., Kuang, W., Lalanne, X., Langlais, B., Léger, J.-M., Lesur, V., Lowes, F.J., Macmillan, S., Mandea, M., Manoj, C., Maus, S., Olsen, N., Petrov, V., Ridley, V., Rother, M., Sabaka, T.J., Saturnino, D., Schachtschneider, R., Sirol, O., Tangborn, A., Thomson, A., Tøffner-Clausen, L., Vigneron, P., Wardinski, I., Zvereva, T., 2015. International geomagnetic reference field: the 12th generation. Earth, Planets Sp. 67, 79. http://dx.doi.org/10.1186/s40623-015-0228-9.

Tucker, D.H., Boyd, D.M., 1987. Dykes of Australia detected by airborne magnetic surveys. In: Halls, H.C., Fahrig, W.F. (Eds.), Mafic Dyke Swarms. Geological Association of Canada, Toronto, ON, pp. 163–172.

Wang, X.-C., Li, Z.-X., Li, J., Pisarevsky, S.A., Wingate, M.T.D., 2014. Genesis of the 1.21 Ga Marnda Moorn large igneous province by plume–lithosphere interaction. Precambr. Res. 241, 85–103. http://dx.doi.org/10.1016/j.precamres.2013.11.008.

Wilde, S.A., Middleton, M.F., Evans, B.J., 1996. Terrane accretion in the southwestern Yilgarn Craton: evidence from a deep seismic crustal profile. Precambr. Res. 78, 179–196. http://dx.doi.org/10.1016/0301-9268(95)00077-1.

Williams, G.E., Schmidt, P.W., Clark, D.A., 2004. Palaeomagnetism of iron-formation from the late Palaeoproterozoic Frere Formation, Earaheedy Basin, Western Australia: palaeogeographic and tectonic implications. Precambr. Res. 128, 367–383. http://dx.doi.org/10.1016/j.precamres.2003.09.008.

Wingate, M.T.D., Giddings, J.W., 2000. Age and palaeomagnetism of the Mundine Well dyke swarm, Western Australia: implications for an Australia-Laurentia connection at 755 Ma. Precambr. Res. 100, 335–357. http://dx.doi.org/10.1016/S0301-9268(99)00080-7.

Wingate, M.T.D., Pidgeon, R.T. 2005. The Marnda Moorn LIP, A late Mesoproterozoic Large Igneous Province in the Yilgarn Craton, Western Australia July2005 LIP of the Month. Large Igneous Provinces Commission, International Association of Volcanology and Chemistry of the Earth's Interia. http://www.largeigneousprovinces.org/05jul.

Wingate, M.T.D., Pirajno, F., Morris, P.A., 2004. Warakurna large igneous province: A new Mesoproterozoic large igneous province in west-central Australia. Geology 32, 105. http://dx.doi.org/10.1130/G20171.1.

Wingate, M.T.D., Pisarevsky, S.A., Evans, D.A.D., 2002. Rodinia connections between Australia and Laurentia: no SWEAT, no AUSWUS? Terra Nov. 14, 121–128. http://dx.doi.org/10.1046/j.1365-3121.2002.00401.x.

Zhai, Y., Halls, H.C., Bates, M.P., 1994. Multiple episodes of dike emplacement along the northwestern margin of the Superior Province. Manitoba. J. Geophys. Res. Solid Earth 99, 21717–21732. http://dx.doi.org/10.1029/94JB01851.

Zhang, S., Li, Z.X., Evans, D.A.D., Wu, H., Li, H., Dong, J., 2012. Pre-Rodinia supercontinent Nuna shaping up: A global synthesis with new paleomagnetic results from North China. Earth Planet. Sci. Lett. 353–354, 145–155. http://dx.doi.org/10.1016/j.epsl.2012.07.034.

Zijderveld, J., 1967. demagnetization of rocks: analysis of results. In: Collinson, D.W., Creer, K.M., Runcorn, S.K. (Eds.), Methods in Paleomagnetism. Elsevier, Amsterdam, pp. 254–286.